

CAN Transceiver Design and Analysis – Project Overview

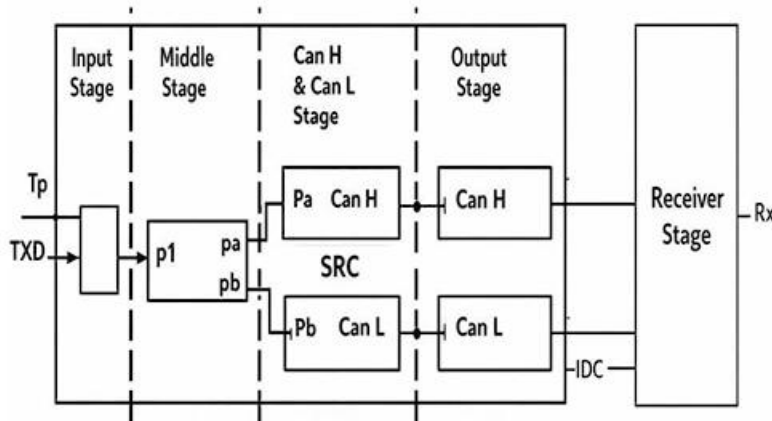


Figure 1: CAN Transceiver Architecture

Description

This figure presents the overall architecture of the CAN transceiver. The design consists of transmitter and receiver sections that interface between the CAN controller and the communication bus. The transmitter converts digital TXD signals into differential CANH and CANL outputs, while the receiver reconstructs the RXD signal from the bus voltages.

Key Highlights

- Differential communication architecture
 - Noise-resistant signal transmission
 - Reliable data recovery
 - Suitable for automotive and embedded applications
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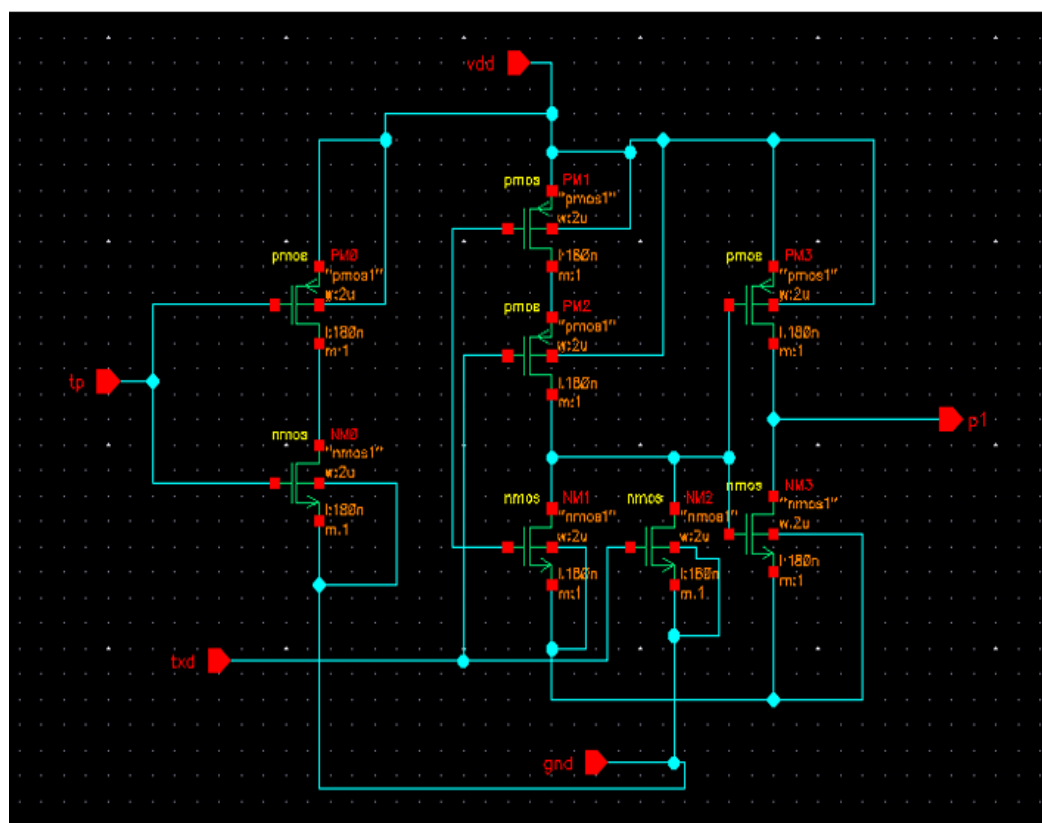
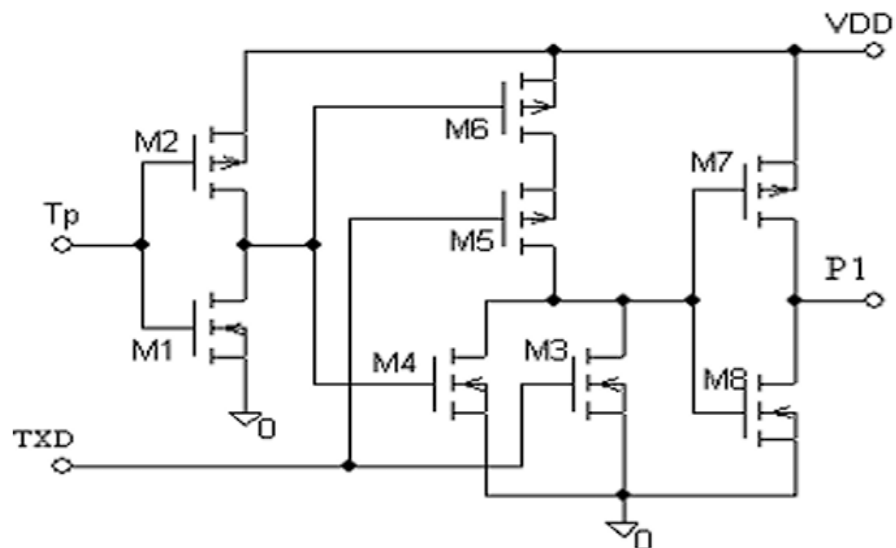


Figure 2: Input Stage Circuit

Description

The input stage receives the TXD signal from the CAN controller and performs signal conditioning. This stage ensures that the incoming signal is properly prepared before being processed by the driver circuitry.

Purpose

- Signal conditioning
 - Stable logic transition
 - Interface between controller and transceiver
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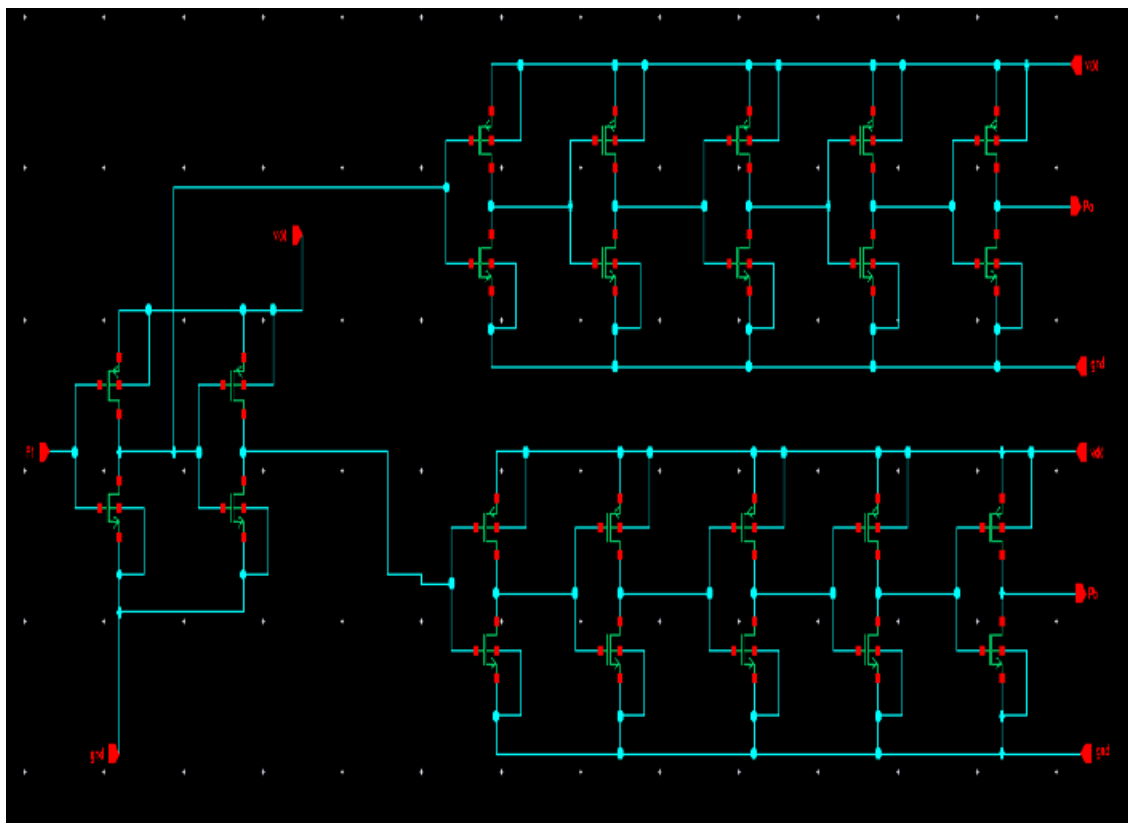
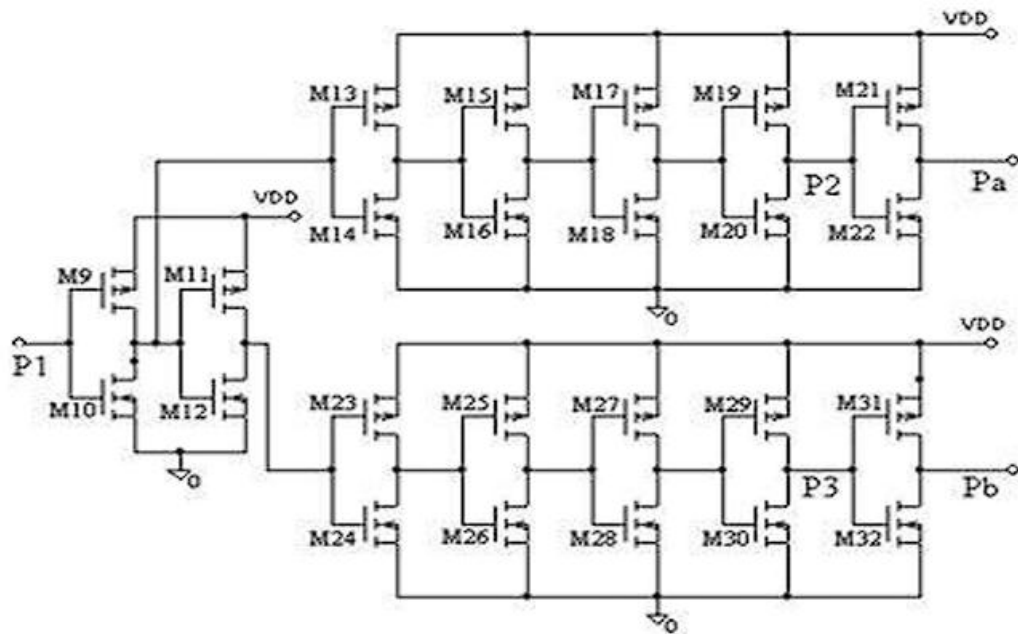


Figure 3: Driver Stage Design

Description

The driver stage amplifies the conditioned signal and provides sufficient current-driving capability required for CAN bus communication. This stage improves switching performance and supports reliable differential signaling.

Purpose

- Signal amplification
 - Improved drive strength
 - Faster switching characteristics
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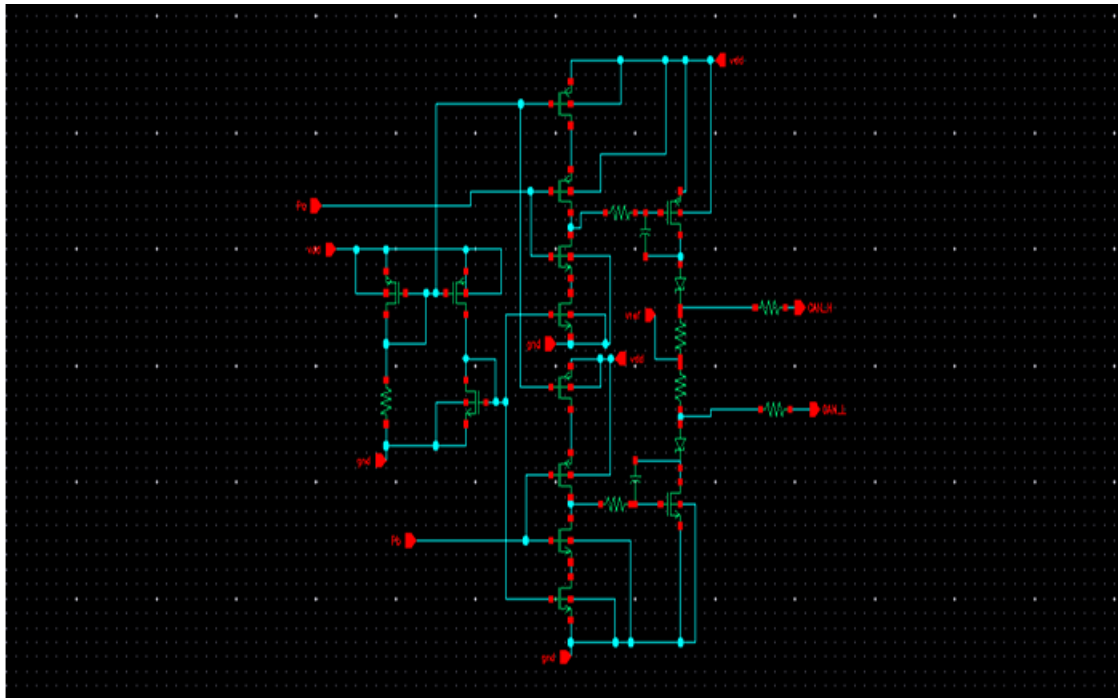
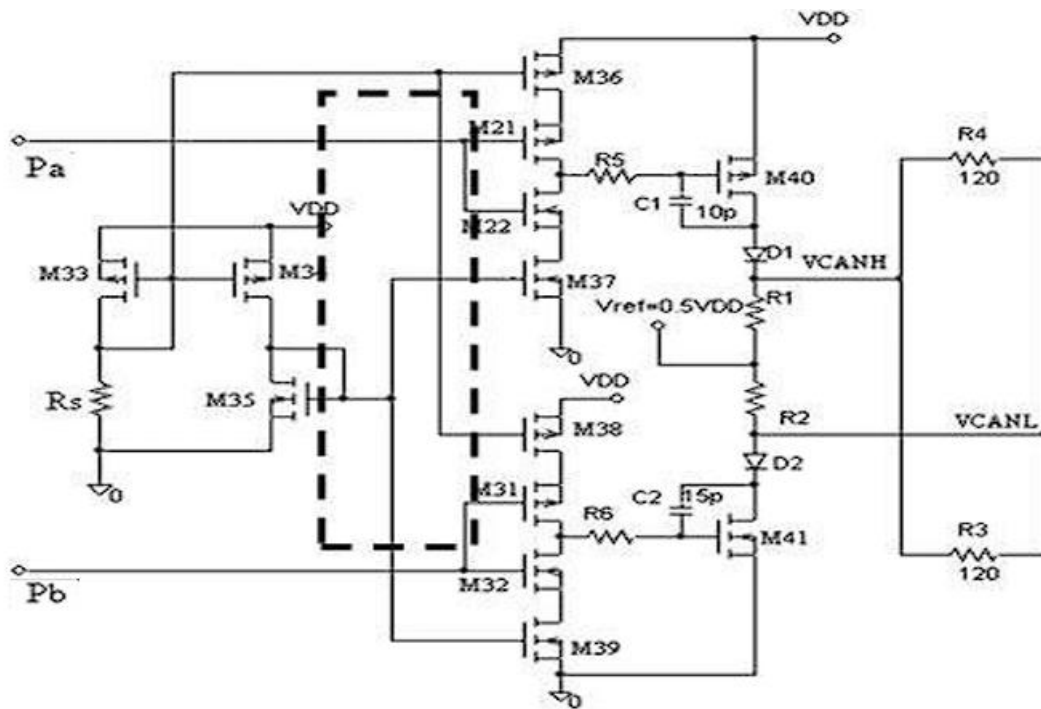


Figure 4: CANH and CANL Output Stage

Description

The output stage generates the CANH and CANL differential signals used for communication over the CAN bus. Information is transmitted through the voltage difference between these two lines, making the system highly resistant to electrical noise.

Purpose

- Differential signal generation
 - CAN bus communication
 - Noise immunity improvement
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Description

The receiver continuously monitors the voltage difference between CANH and CANL. It reconstructs the original digital signal and provides it to the CAN controller through the RXD output.

Purpose

- Differential signal detection
 - Noise rejection
 - Reliable data recovery
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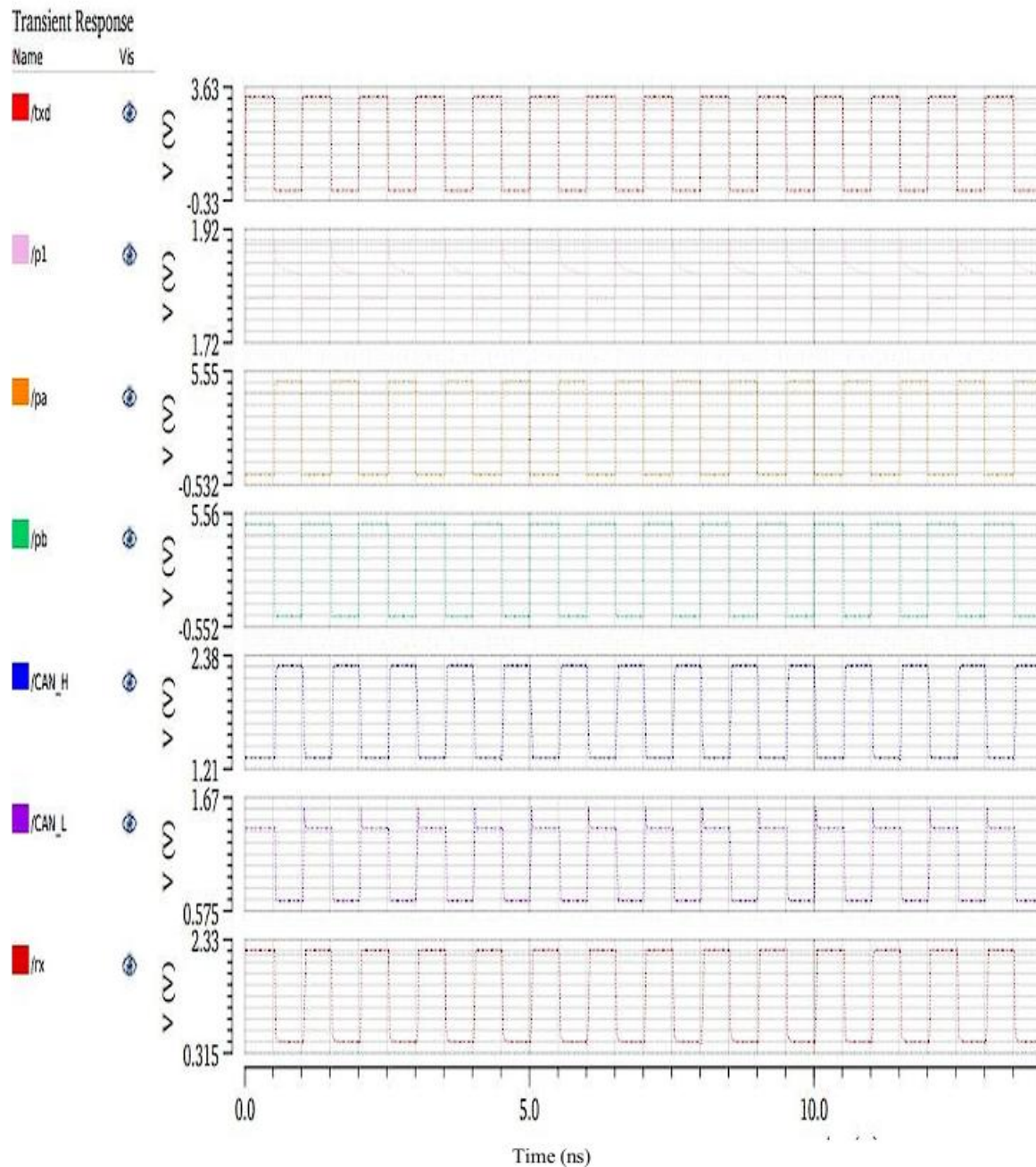


Figure 6: Simulation Results

Description

The simulation waveform verifies the operation of the CAN transceiver. The generated CANH and CANL signals follow the expected differential communication behavior, while the receiver correctly reconstructs the transmitted data.

Observations

- Stable differential signaling
 - Successful transmitter operation
 - Reliable receiver functionality
 - Correct data reconstruction
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Project Outcome

The project successfully demonstrates the design and analysis of a CMOS-based CAN transceiver using Cadence tools. The transmitter and receiver circuits were verified through simulation, confirming reliable differential communication suitable for automotive and embedded system applications.

Skills Demonstrated

- CMOS Circuit Design
- Cadence Virtuoso
- Verilog HDL
- VLSI Design
- Communication Protocols
- Simulation and Verification